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TỰ ĐỘNG HÓA-THÔNG MINH HÓA

COMPREHENSIVE AUTOMATIC SOLUTION
FOR FURNITURE MANUFACTURING

BAZIS

AUTOMATIZATION - INTELLIGENTIZATION

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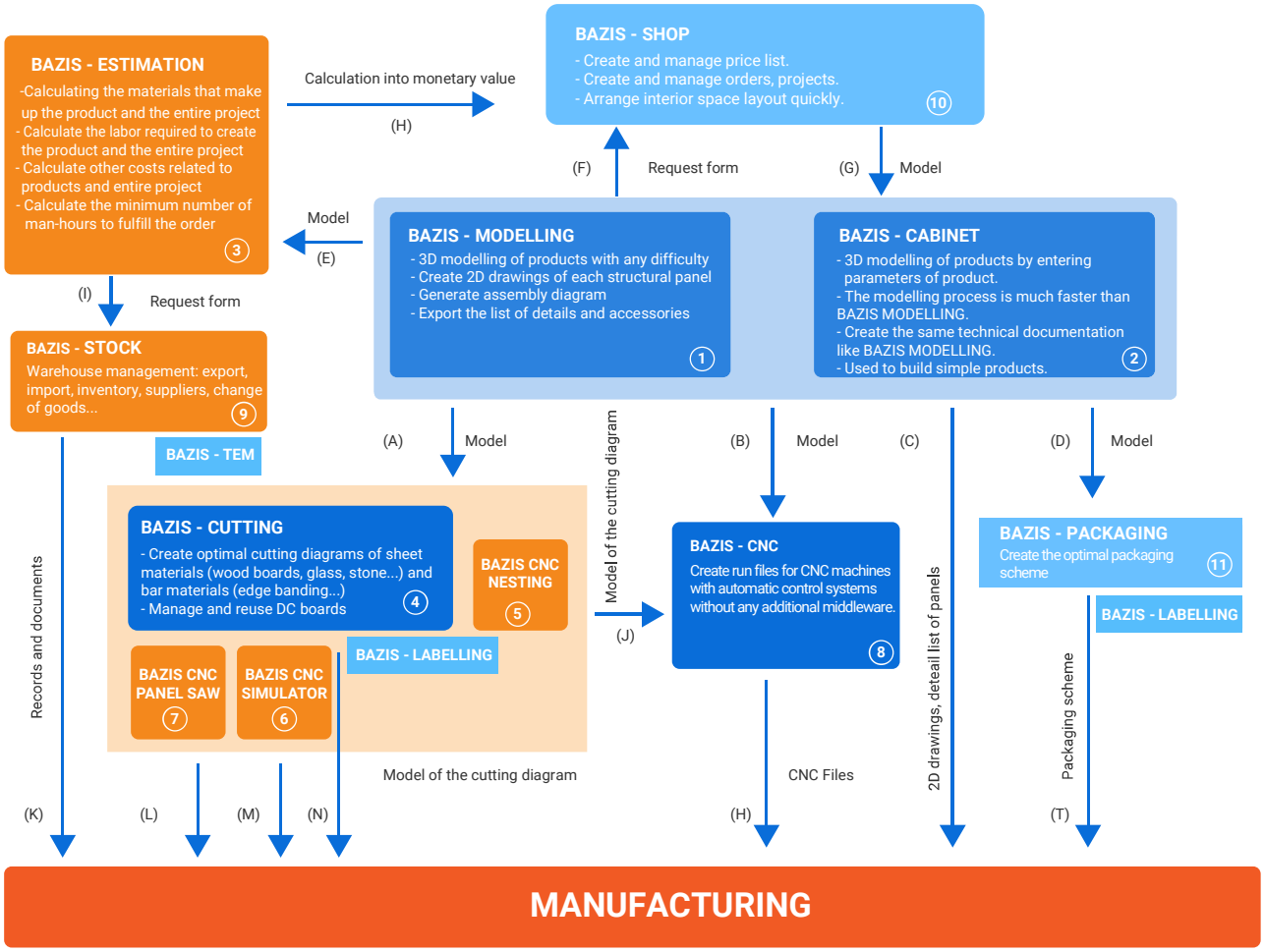
BAZIS – COMPREHENSIVE SOLUTION
FOR FURNITURE MANUFACTURING

BAZIS is a complete system of software modules. Each module can operate independently, as well as interact with other modules in a unified information environment. The modular structure of the BAZIS system helps manufacturers to perform each step in the production process in a rational way, making optimal use of raw materials and energy.

The BAZIS system will achieve maximum efficiency when combining flexibly between the modules in the system.

OVERALL SYSTEM STRUCTURE OF BAZIS



Outstanding advantages

BAZIS - is a software system, customized and localized by WOODSOFT VIETNAM (hereinafter referred to as WOODSOFT). BAZIS is applied in hundreds of large and small production units in Vietnam (Hanoi, Bac Ninh, Hung Yen, Quang Ninh, Hai Phong, Nghe An, Da Nang, Buon Ma Thuot, Binh Duong, Dong Nai, Tay Ninh, Ho Chi Minh, etc.).

The BAZIS system has a great advantage over European, American and Chinese software because it has Vietnamese experts to directly support and quickly handle customer problems in the manufacturing process.

BAZIS - is a specific solution for the furniture industry in Vietnam.

BAZIS software system provides furniture manufacturing companies all the necessary tools to: manage inventory of supplies and accessories, design and modelling products, optimize cutting boards, integrate with processing machines (CNC, Panel saw, etc.), building a unified information environment of the enterprise.

One of the most outstanding advantages of BAZIS is its compatibility with all machines using in the furniture manufacturing industry in Vietnam, including CNC machines, Panel saws... from China, Italy, Germany, etc...

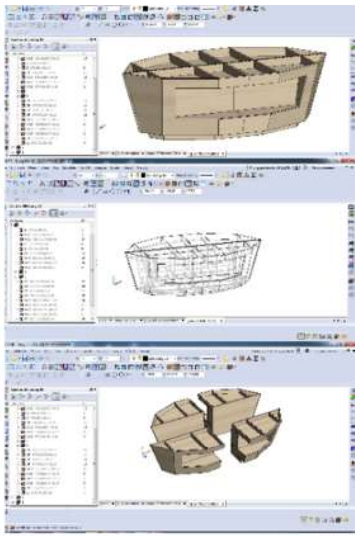
BAZIS also brings benefit to enterprise production system through building and controlling BOM (Bill of Materials) and BOO (Bill of Operation) databases.

1 BAZIS MODELLING

The most important module, the heart of the BAZIS software system is the BAZIS MODELLING module. The BAZIS MODELLING module is the ideal combination of operational simplicity with unlimited possibilities in product modelling.

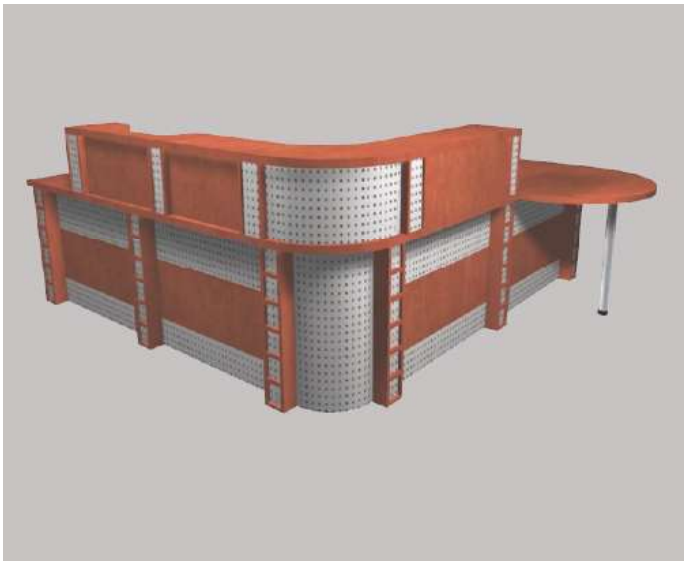
With the BAZIS MODELLING module, designers can create wooden products of any complexity. Ability to generate detailed 2D drawings of each component that makes up the product, detailed statistics of boards, detailed drawings of projections, sections, and perspective of the product automatically. The product's assembly schema is also optimized and generated by the software's algorithm.

Here are some products designed by BAZIS MODELLING and actual images:



The outstanding advantages of BAZIS MODELLING:

SIMPLE DESIGN PROCESS

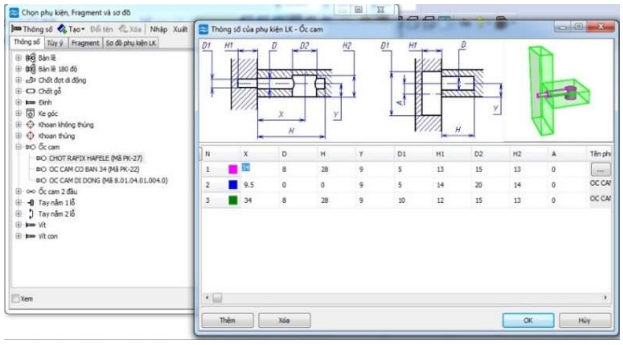


The product design process is extremely simple, straightforward and intuitive. The product designer acts as a carpenter on the computer.

The process of placing parts in the drawing is done in one of three ways: from the front, from the left or from the top.

This method is very intuitive and simple for any designer, as the design process is in fact the creation of an assembly schema in space.

CONVENIENT INSTALLATION OF CONNECTION ACCESSORIES



The types of connection accessories in the BAZIS system are extremely diverse, divided into specific accessory groups such as connecting bolt, wooden dowels, connecting screws, confimat, corner connection accessories, movable connecting screws, etc.

To add a new type of connection accessory, the user just needs to enter the parameter in the corresponding accessory group. Using these accessories is simple: just indicate the parts to be joined and the position of the connection accessories along the length of the joint. Alternatively, the second way is to set a fixed interval and the accessories will be placed automatically according to the length of the part.

If the BAZIS CABINET module is used for design, the connection accessories will be automatically arranged into the product details in an optimal way.

CONVENIENT EDITING

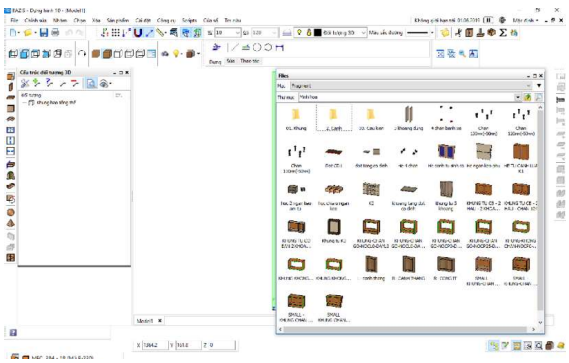
The BAZIS MODELLING module contains a set of commands that allow the user to easily and quickly obtain a new product from a previously designed product model. This allows users to significantly save time when designing a typical model of furniture and creating a wide range of variations from it.

This combination of commands has a function that allows performing operations such as: change the packaging size of the product, replace the original materials and components with another type... It also allows the removal of the product, remove, modify or add any product details.

DESIGN AND USE OF TYPICAL PARTS (FRAGMENT)

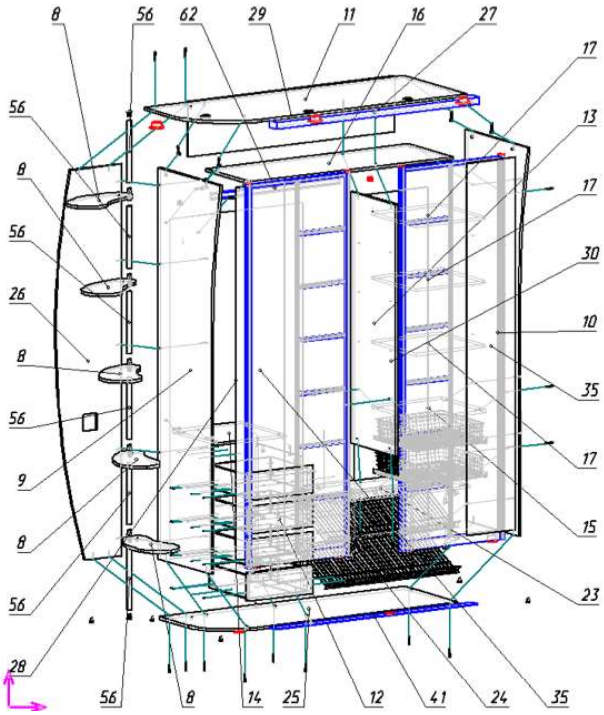
When working on the BAZIS System, users do not need to redesign a large number of frequently using parts (sample or standard products, accessories, etc.)

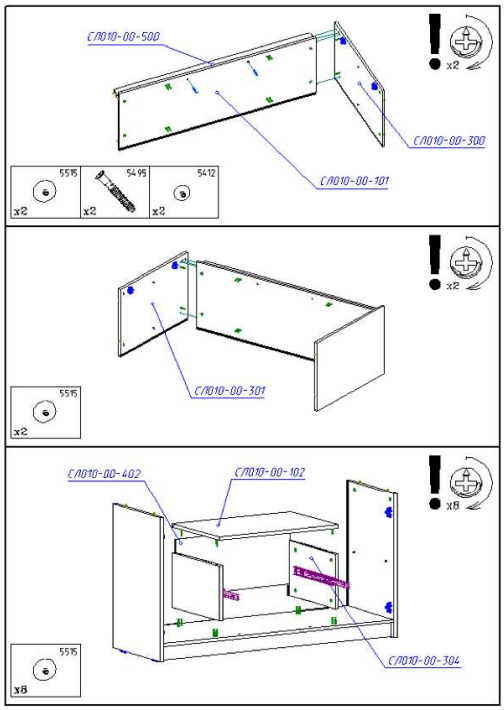
In order not to waste time creating duplicate parts, users can simply create



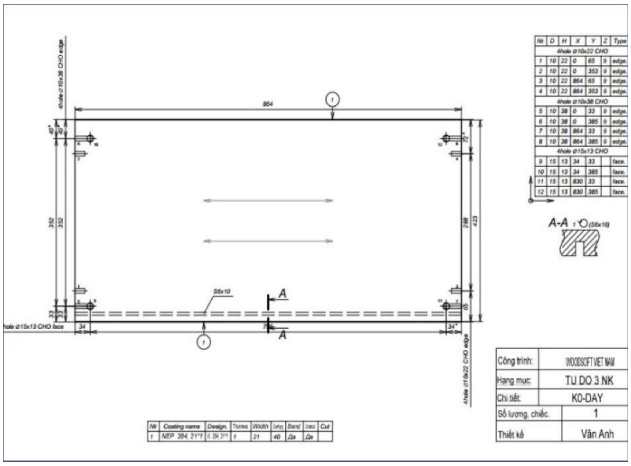
that part once, save it in the system. Then if it needs to be reused, just put that existing design in any position of the product structure.

AUTOMATICALLY CREATE TECHNICAL DRAWINGS





This function frees up the designer's "classic work", allowing them to simply edit the drawings as needed.



The BAZIS system will make fully automatic and complete assembly instruction schema according to the “Design Standards”, as well as assembly drawings for each detail. Accompanied by the necessary technical parameters.

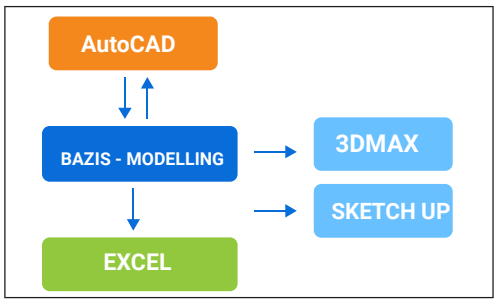
When publishing a drawing, many settings that allow the user to specify the drawing specification. All drawings created automatically on the system are ready for immediate use in production.

COMPATIBLE WITH OTHER SOFTWARE

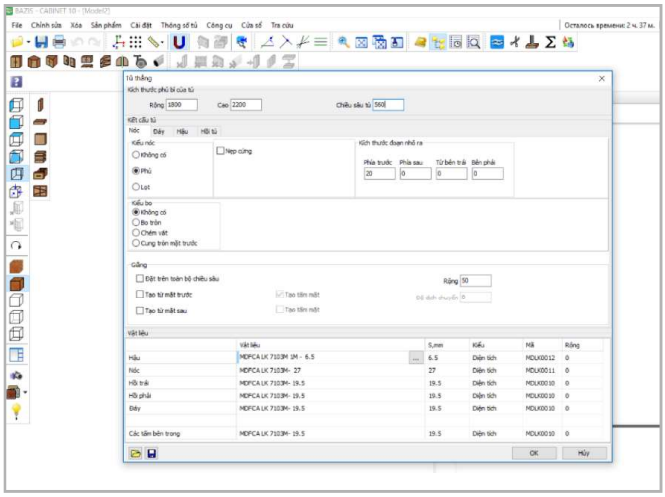
BAZIS MODELLING allows:

- Import 2D drawing data from AutoCAD software, to inherit sketch drawings, Vertical view, side view, horizontal view from existing AutoCAD drawings.
- Import 3D drawings from Sketchup to inherit the work of the design department.

- Export to other 3D software such as: 3D MAX, Sketchup...
- Export modelling data to other formats such as .dbf, .xml, .xls...



2 BAZIS CABINET

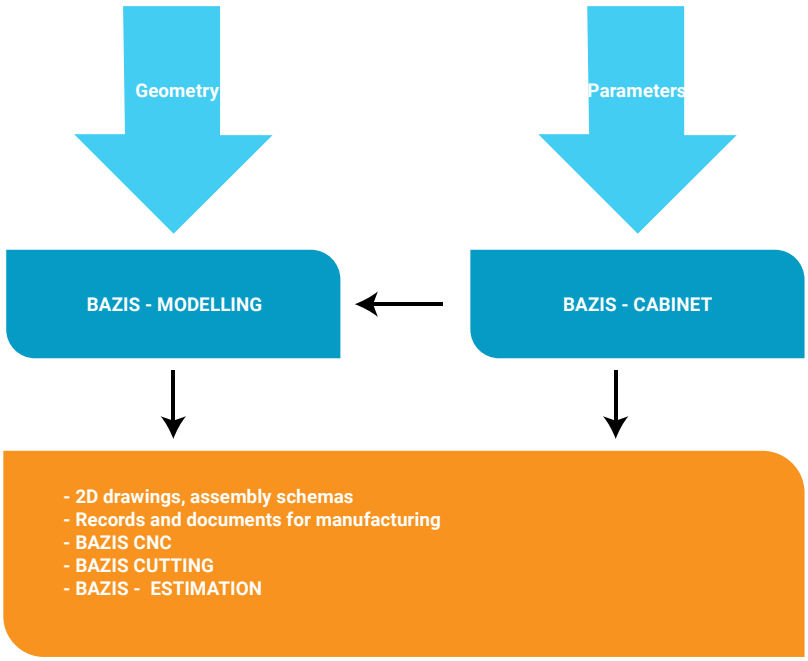


BAZIS CABINET allows fast and accurate modelling of all cabinet furniture. Allows working with parametric 3D objects, reducing the time to create new products.

Users can edit the specifications and save the product's parameters as well as the product's internal structure for reuse for later modelling.

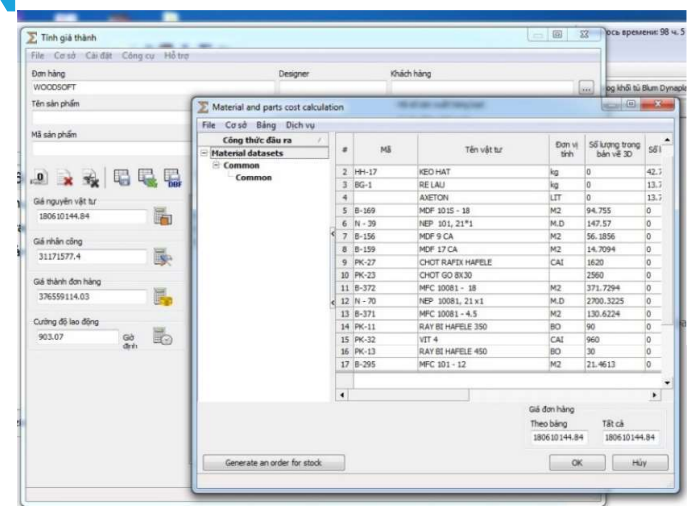
Unlike the basic BAZIS MODELLING module, BAZIS CABINET does not require the designers to have a lot of technical knowledge.

BAZIS CABINET allows:
- Combine with BAZIS MODELLING module to edit products created from BAZIS CABINET module.
- Check errors when modelling: details are oversized board, cannot open drawer because of entangling in cabinet door...



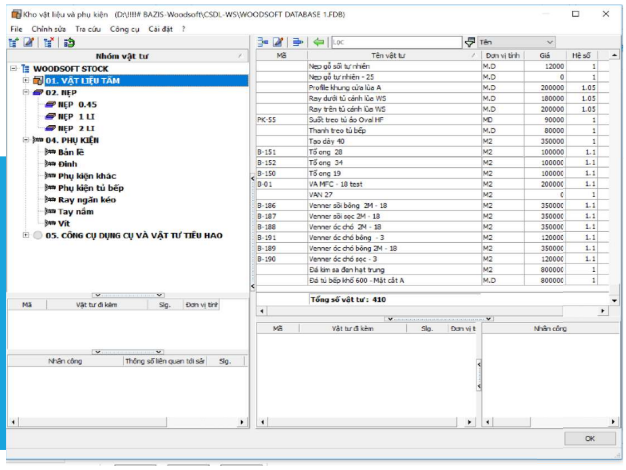
Input data
For BAZIS - MODELLING: basic shapes
For BAZIS - CABINET: Change, select pre-programmed parameters
Output data: The same

3 BAZIS ESTIMATION



BAZIS ESTIMATION allows to automatically calculating the production cost of furniture products (including Cost of materials and cost of labor). It also allows setting up the formula to calculate the cost of products and works in a flexible way, close to reality.

From BAZIS ESTIMATION, users can export BOM (Bill of Materials) and BOO (Bill of Operation) data for synchronization with enterprise ERP systems.



BAZIS ESTIMATION manages the enterprise's database of materials and labor.

Material data can be imported directly into the software through ".dbf" or ".xml" format.

Material and labor norms are entered directly into the software to monitor and optimize the use of materials and worker productivity.

The formula for calculating the price is set up basing on historical statistical data of materials and labor expenses for each product type.

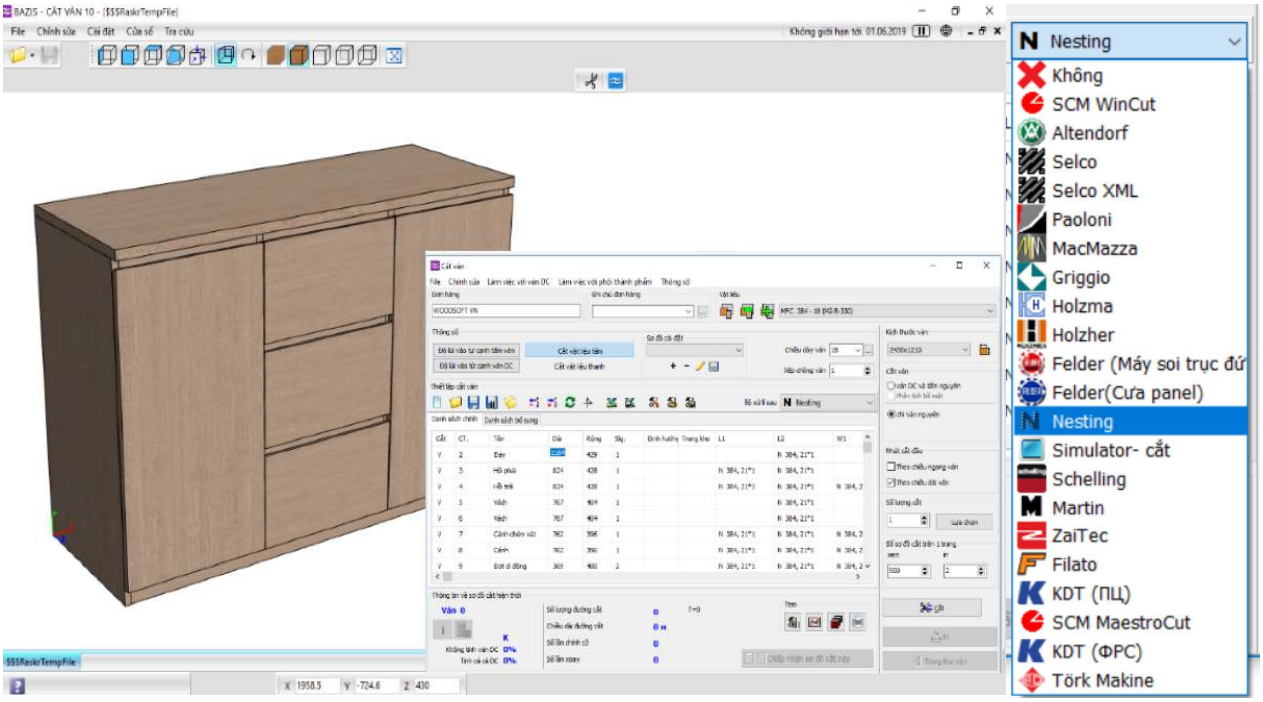
4 BAZIS CUTTING

BAZIS - CUTTING significantly saves materials and manufacturing time because the cutting plan is optimally calculated and displayed as an intuitive, easy-to-understand chart.

Optimizing the board cutting scheme by many intelligent algorithms. The priority criteria and the cutting order are changed specially for each machine type: Sliding Panel saw, CNC Panel saw, Nesting CNC.

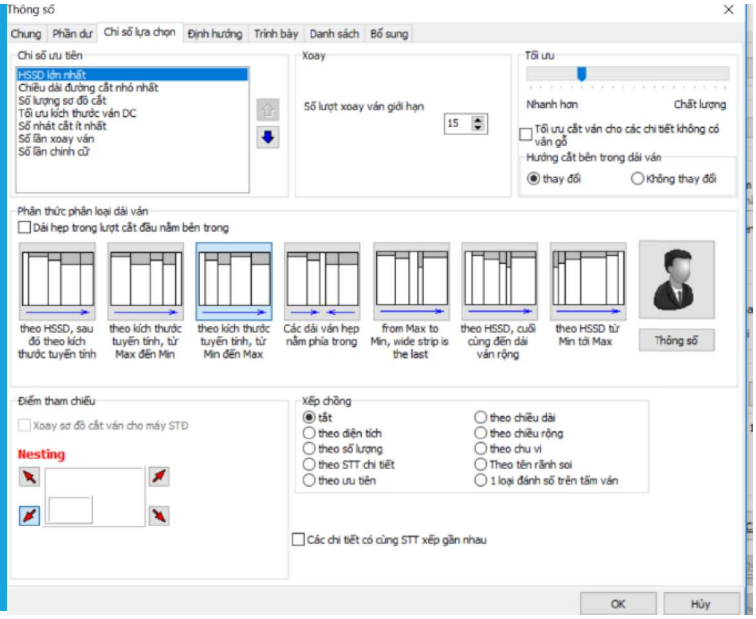
Possibility to filter panels by border, size, defined or undefined texture's orientation.

It is possible to create cutting plan not only for one product but also for a multi-product project, each product can have different quantities.



The included label design and printing module allows designing labels according to the needs of any enterprise.

Label templates can be flexibly customized for each enterprise.

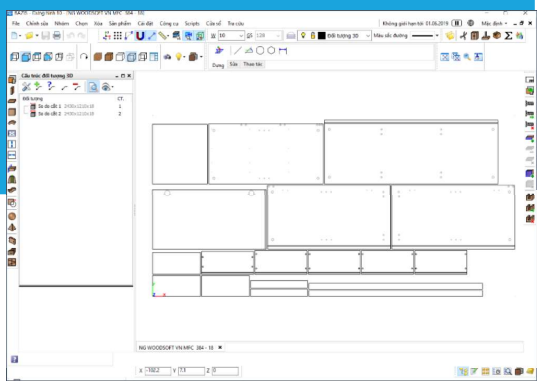


Information fields in the label-printing module can be added and further set up based on: basic parameters, complex parameters and fixed parameters.

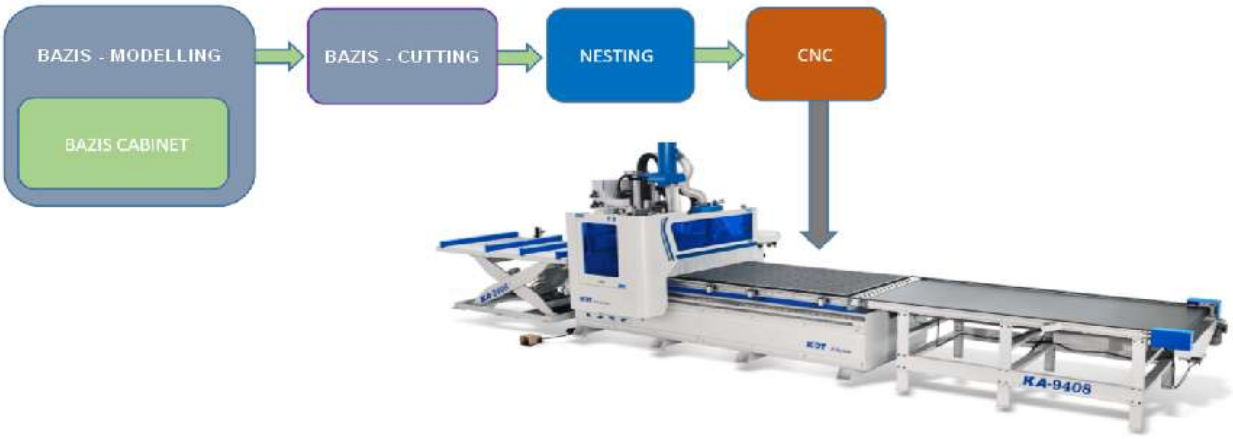
Label templates can be flexibly designed according to the needs of each company and can be printed on different paper sizes.

5 BAZIS - NESTING

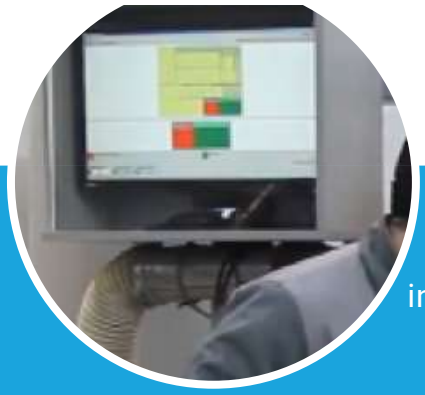
BAZIS – NESTING is a special postprocessor that acts as an intermediate cutting plan with 3D drill holes to output to central Nesting CNC machines through BAZIS CNC.



Normally, central Nesting CNC machines run according to the following cycle:



6 BAZIS – SIMULATOR



BAZIS SIMULATOR is a type of CNC simulation for a sliding panel saw, in which all operations performed on a sliding panel saw are described in the same detail as on an automatic panel saw.

Simulator is also one of many post processors of BAZIS CUTTING.

7 BAZIS CNC - PANEL SAW

BAZIS CNC - PANEL SAW is one of the post processors of BAZIS CUTTING especially for automatic panel saws.

The post processor of the BAZIS - CUTTING module generates cutting plans and control programs for the following machines:

	Machine control system	File formats
Altendorf	Wincut	*.saw
Casadei Macchine (SCM Group)	for saw machines	*.scm (by default)
Felder	for panel sizing saws	*.may
Filato	Filato	*.
Gabbiani (SCM Group)	Wincut	*.csv
Griggio	iLENIA	*.scm (by default)
HOLZ-HER	HOLZ-HER	*.nps
Holzma (HOMAG Group)	Postprocessor	*.ncr
KDT	HolzLink (HOMAG Group) is necessary	*.ptx
MacMazza	KDT	*.txt
Martin	-	*.pro
Paoloni	Martin	*.mrt
Schelling	-	*.txt
SCM (SCM Group)	-	*.ncl, *.ncp
Selco (BIESSE Group) до 2000 г.	Wincut	*.scm (by default)
Selco (BIESSE Group) после 2000 г.	NC	*.
ZaiTec	Needed postprocessor XML-Link (Biesse Group)	*.xml
	ZaiTec	*.txt
	Simulator	*.sim
	Nesting technology	*.b3d

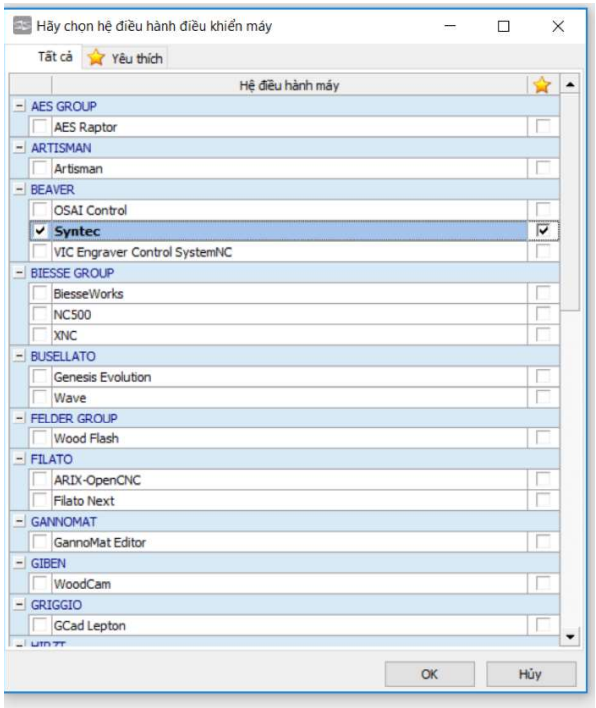
Note: the list of machines compatible with BAZIS is regularly updated. If you need to check the compatibility of the machines, please contact Woodsoft for support.

8 BAZIS – CNC

BAZIS – CNC is a middleware that converts from 3D drawings on BAZIS MODELLING or BAZIS CABINET modules as well as nesting diagrams to the appropriate format of CNC machines.

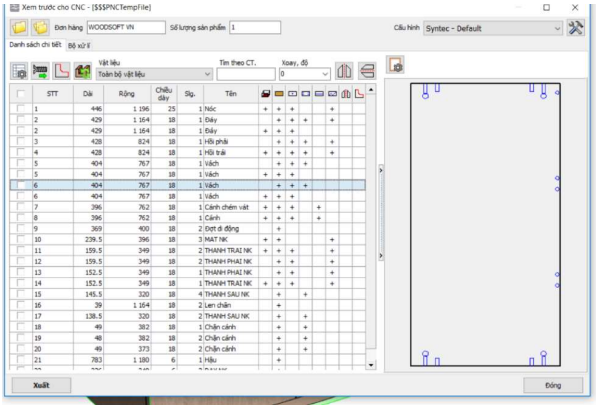
Automatically receive detailed information from 3D objects as well as 2D drawings (CAD) to export to the required format of CNC machines.

Automatically create combinations of commands to process surface drilling, surface milling, edge milling, and edge drilling....



Automatically generate two drawings from the main and the other side, allowing users to specify machining operations for convenience in the manufacturing process.

Exported to a file format in which parts are banded or not.



User can set the file to run for multiple formats at the same time.

Set the machine to mill slots using multiple tools.

Check the feasibility of the steps on the machine.

Optimizing the drilling process.

CNC machines can be directly connected to BAZIS

Vendor	Machine type	Operation System	Format
Artisman	KP	Artisman	NC
Beaver	KP	VIC Engraver Control System	NC
Beaver	KP	OSAI Control	CNC
Beaver	KP	Syntec	CNC
Biesse Group	KP	NC500	DXF, PAN
Biesse Group	KP	NC1000, BiesseWorks Editor, XNC	BPP, *.
Busellato	KP	Genesis Evolution 6.5-8.8	CNC
Busellato	KP	Wave (Xilog Maestro)	XXL, PGM
FELDER	KP	Genesis Evolution 6.5-8.8	CNC
seri Format4 Profit 2s	KP	Wood Flash	TCN
FELDER	K	ARIX-OpenCNC	TNC
seri Format4 Profit H20-30	K	Filato Next (NC Boring)	AUT
Filato	K	GannoMat Editor	ASCII
Filato	KP	WoodCam	DXF
GANNOMAT	K	GCad Lepton	PZA
GIBEN	KP	TpaCAD Albatros	TCN
Griggio	K	iLENIA CAD/4	PD4
Griggio	KP	Campus Hops	HOP
Hirzt	KP	TwinCam	DXF
HOLZ-HER	KP	WoodWop 4.5-6.1	MPR
HOLZ-HER	KP	ImaWop 4.0-8.0	FMC
Homag Group	K	NC-Studio	NCP
IMA (BIMA)	K	ICE101	*.
Koch	K	iLENIA CAD/4	PD4
Maggi	K	MW208	LBR, UTE
Maggi	KP	MW310	DXF, TLF
MasterWood	KP	MultiCam PSS	NC
MasterWood	KP	Xilog Plus , Xilog Maestro	XXL, PGM
MultiCam	B	Cyflex H800	CNC
SCM	KP	Albatros EdiCAD	*.
SCM	K	VitapPoint CAD/4	PD4
Uniteam	K	Albatros (Wood Flash)	PD4
VITAP	KP	VIC Engraver Control System	CNC
VITAP	KP		DNC
Zelder			
Артель			
KP – milling and drilling		K - drilling only	

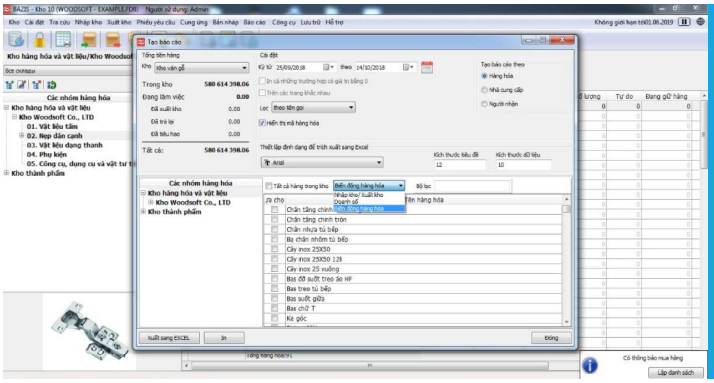
Note: the list of machines compatible with BAZIS is regularly updated. If you need to check the compatibility of the machines, please contact Woodsoft for support.

9 BAZIS STOCK

BAZIS STOCK makes storing and controlling materials easier. It also makes the supply process smooth and convenient.

Using BAZIS – STOCK helps optimize the work of the design department, the supply department. It supports continuous production with all the necessary supplies and tools.

Features of BAZIS STOCK:



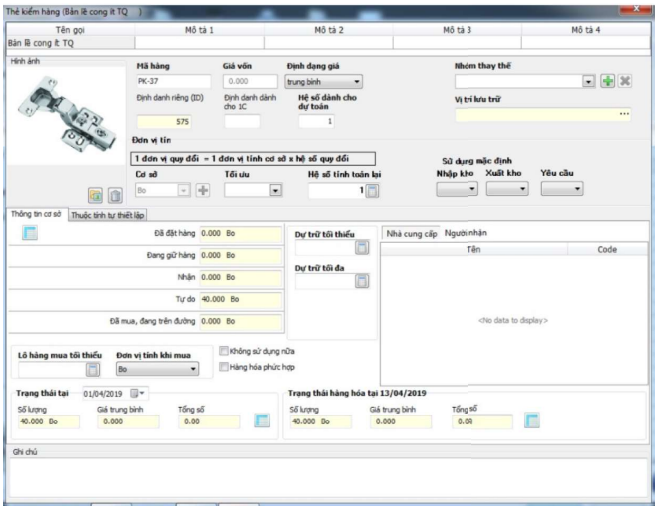
- Design warehouse hierarchical structure with internal product flow.
- The vouchers are generated automatically.
- Can transfer and exchange information with other systems.

- Setup the product's label with the support of BAZIS - LABELLING module.
- Create records and documents through barcode scanning.
- Easy to work with suppliers.
- Easy to use interface helps to save user settings.
- Ability to manage finished product inventory.
- Many users can work with products and orders in the same unified Database.
- Works effectively with Request forms, exported from BAZIS - ESTIMATION.

BAZIS - STOCK cannot replace accounting software, but they complement each other, solving small but extremely important problems for businesses.

The maximum efficiency of BAZIS – STOCK is achieved when it is used in combination with the BAZIS – MODELLING and BAZIS – CABINET modules.

BAZIS - STOCK can also operate separately like a normal stock management software.



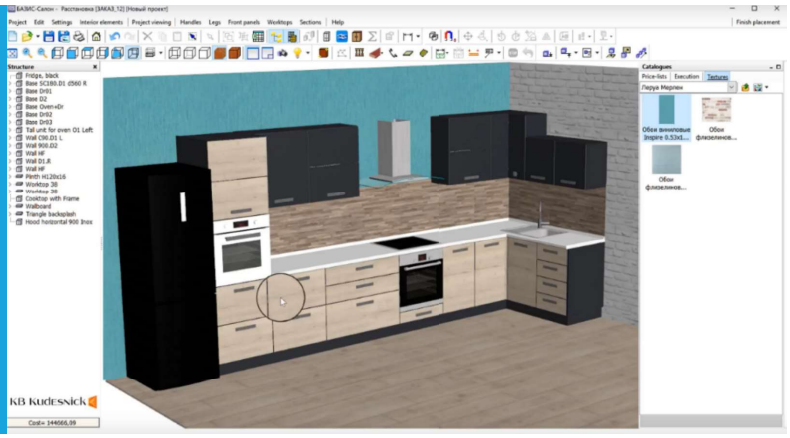
10 BAZIS - SHOP

BAZIS - SHOP quickly design interior spaces, creates orders from both standard and non-standard products.

BAZIS - SHOP has three sub-modules:

- PRICE LIST MANAGEMENT: creates pricelist for furniture modules and standard products.
- ORDER MANAGEMENT: enter and manage orders.
- BAZIS – INTERIOR: Arranges interior space.

Module PRICE LIST MANAGEMENT used to build the catalog of the furniture products. Group products by function and manage their attributes.

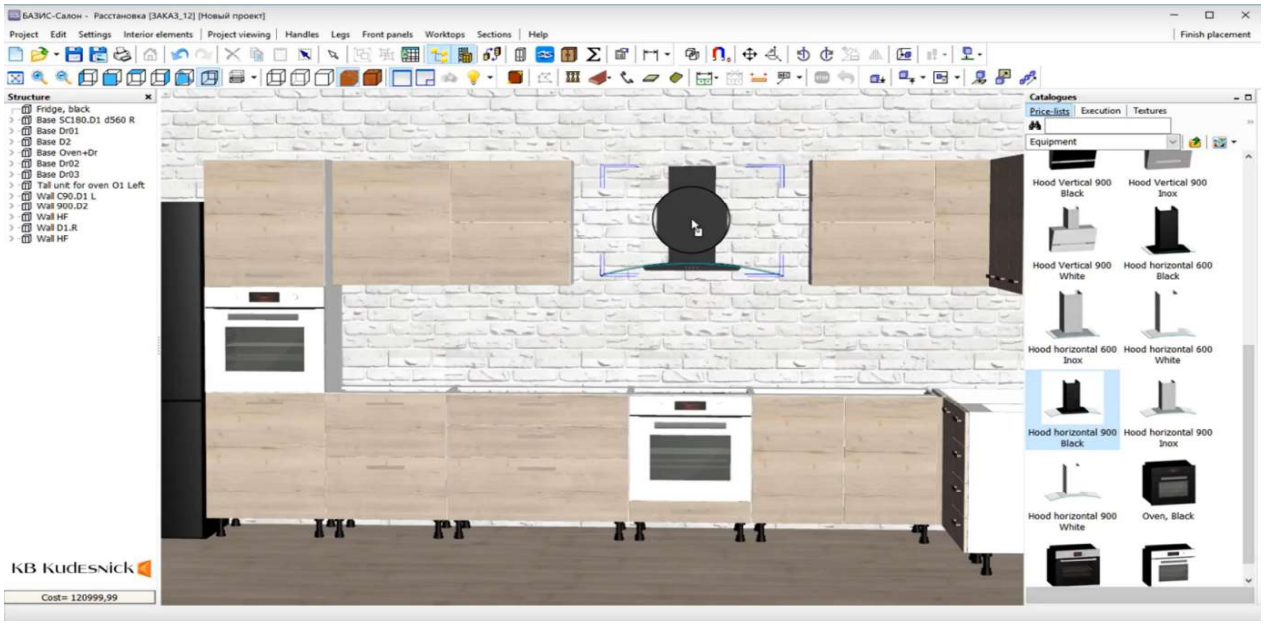


Module ORDER MANAGEMENT:

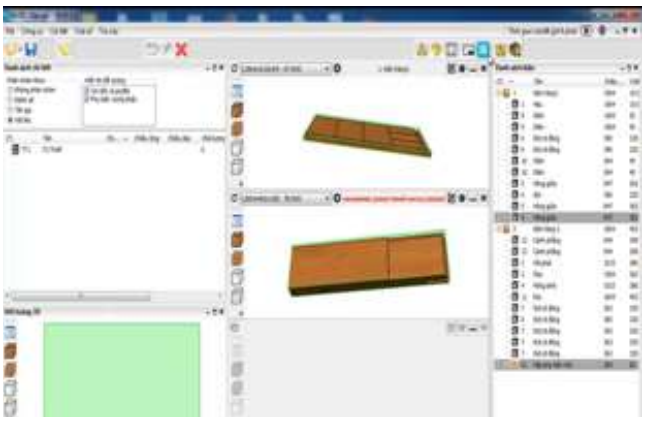
- Used to automate the creation of orders, manually editing if necessary. Helps to manage database-based orders, customer lists, extensive customer services and all phases of customer support (starting from signing contracts, placing orders, designing, manufacturing to shipping products to customers).
- Add products created on BAZIS MODELLING and BAZIS CABINET to the orders with the ability to automatically calculate product costs and adjust costs.
- Create a table of additional services and workers not directly involved in the production process at the factory. Calculating the cost of these services is the same as BAZIS – ESTIMATION.
- Set up templates, text forms (Statistical tables, invoices...) with the support of specialized reporting software FastReport.
- Organize and manage orders created by remote showrooms.
- Edit orders manually, using products from the current price list.
- Report the revenue of each showroom as well as each salesperson.

BAZIS – INTERIOR arranges interior space

- Used to design interior space.
- Quickly build up interior space with the layout of windows, doors, room dividers…
- Automatically replace product models or components such as handles, stands... with other similar accessories from the price list.
- Illustrate the movement mechanism of hinges, sliding doors, drawers...
- Transfer orders to production



11 BAZIS – PACKAGING



- Automatic, easy packing of panels with different sizes.
- BAZIS PAGAGING helps to quickly and accurately place the panels into the box, fitting the package.
- Packages and boxes of screws can be placed in the bales.
- Automatic design of different types of packaging for each bale depending on the allowable weight of the bale.

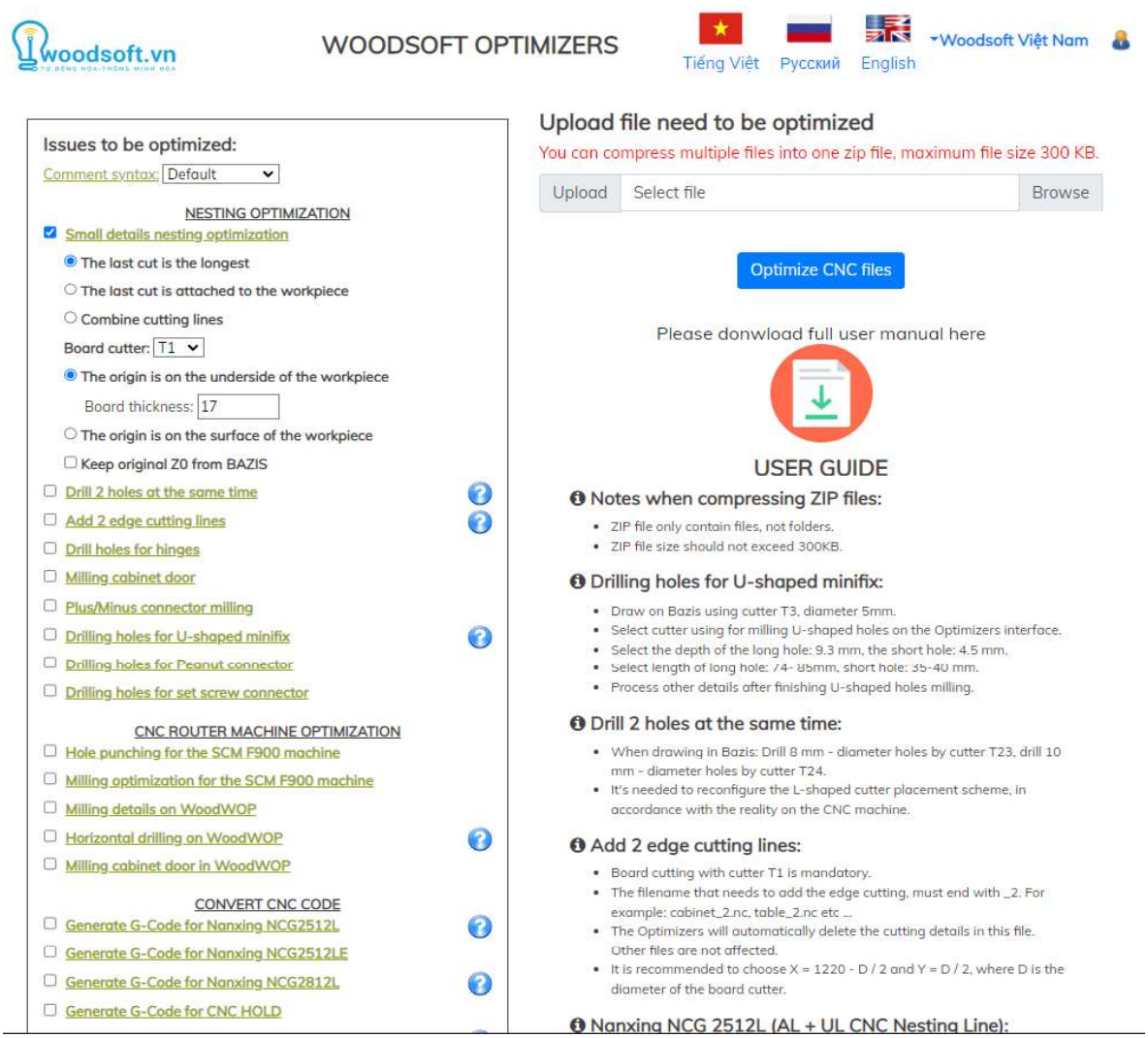
- Automatically create the arrangement schema of details in each package.
- Automatic generate labels for each package

12 WOODSOFT OPTIMIZERS

With the desire to add the experience of Vietnamese carpenter to foreign software, converging and realizing the secrets, tips and ideas of Vietnamese people in the furniture manufacturing process, Woodsoft developed the Woodsoft Optimizers software.

Key features of Woodsoft Optimizers are:

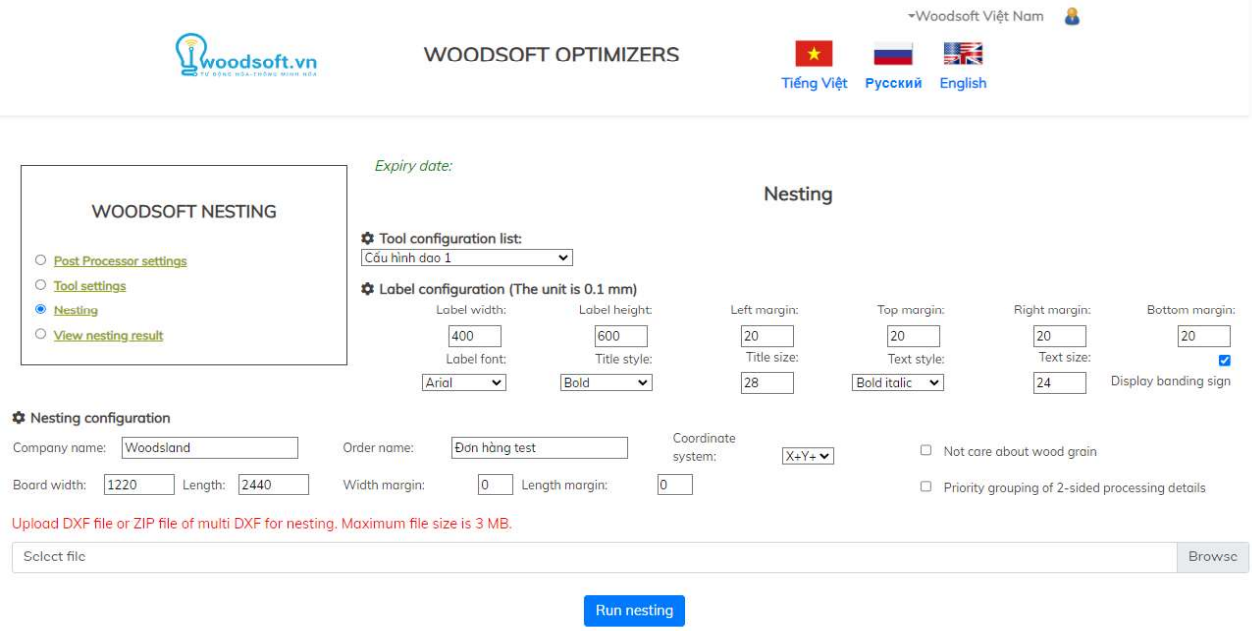
- Optimizing the cutting process: Preventing small details when Nesting, Running 2 holes at the same time, board edge cutting, milling for U-shaped minifix, connecting screws, milling cabinet door, optimal hinge hole drilling etc.
- Optimizing part processing on machining centers: Optimizing part milling, Optimizing horizontal drilling, Optimizing hole drilling, etc.
- Generate processing code for different machines: Nanxing, Huahua, KDT, SCM, etc.
- Import Sketchup file into BAZIS
- Import parameters from Excel file (.csv) for automatic creation of models in BAZIS.



13 WOODSOFT NESTING

Woodsoft Nesting is a software completely developed by Woodsoft with the aim of reducing software costs for customers and simplifying the process of generation running files for CNC machine.

Woodsoft Nesting's philosophy is ALL IN ONE CLICK.

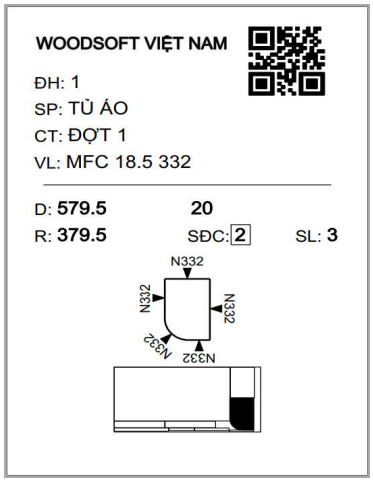


Outstanding features of Woodsoft Nesting:

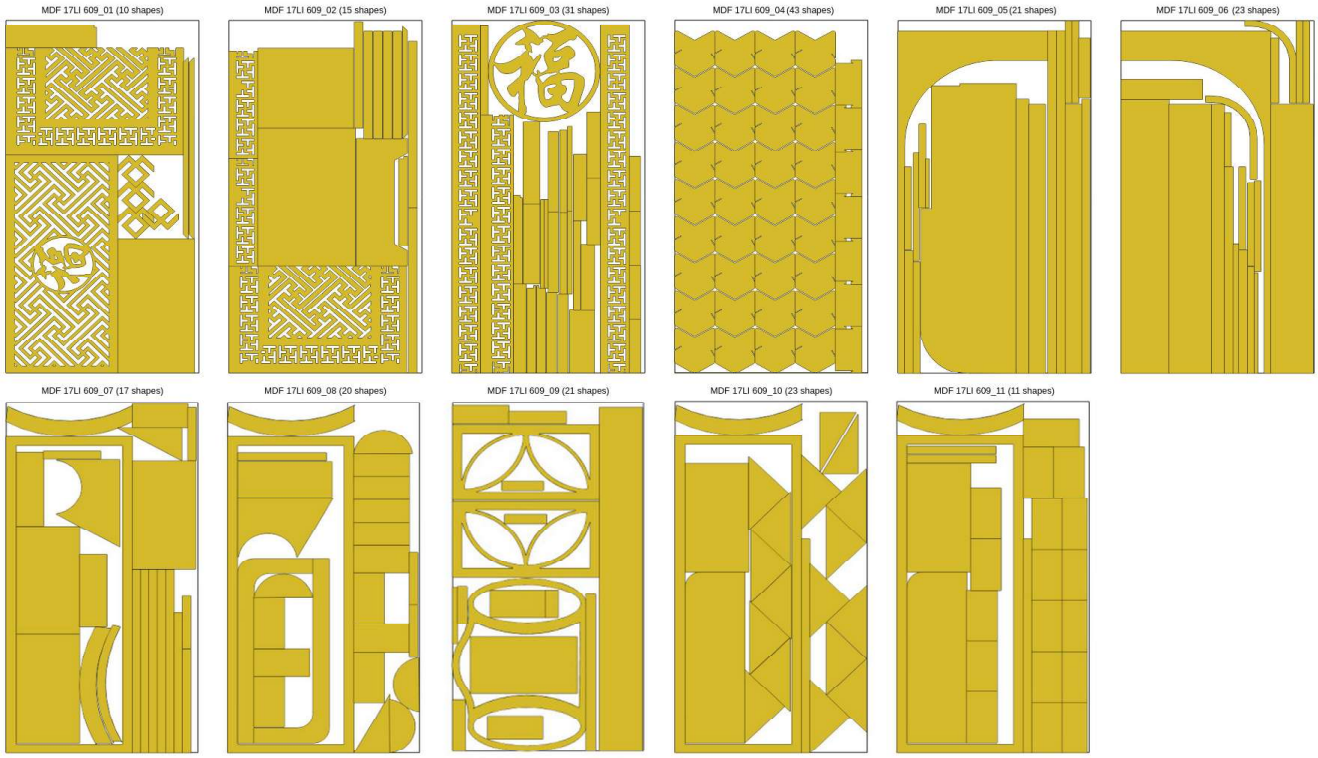
- With just 1 click, users get all nesting results including: CNC running files, labels, reports of consumables, etc.
- Optimal cutting plan, maximum saving of wooden boards, placing one part in another
- The CNC running files already include specific tips to prevent deviation of small details, optimal hinge hole drilling, board edge cutting, milling for U-shaped minifix, connecting screws, milling cabinet door etc.
- There is a lot of useful information on the label: order name, product name, part name, material, size of part, edge banding mark etc.
- Allows reusing the DC boards, generate CNC running files according to the correct size and number of DC boards.

Labels with full information to reduce worker errors:

- Generate beautiful labels, full of information: edge stickers, drills, holes, horizontal drills, etc.
- Marking even curved edge banding.



Woodsoft Nesting generate optimal board cutting plan:



Consumables report:

Detailed statistics report help s optimize material consumption, convenient an easy calculate cost:

- Each type of board material.
- Each type of edge banding.
- Quantity of each main accessory: hinges, cam screws...
- Statistics of milling slot types.

woodsoft.vn TỰ ĐỘNG HÓA · THÔNG MINH HÓA	**WOODSOFT NESTING** https://woodsoft.vn/optimizers/nesting Thời gian nesting: 19:49:18 04/04/2023

Danh sách vật liệu

#	Vật liệu	Kích thước	Số lượng ván	Số lượng chi tiết	Lượng nẹp
1	MDF-_389SL-_17mm	2440.0 x 1220.0 x 17.2	9	97	N1X44-389: 10.21 m (Tiêu hao: 10.49 m) N1X21-389: 213.72 m (Tiêu hao: 229.52 m)
2	MDF-431BT_-_17mm	2440.0 x 1220.0 x 17.2	3	5	NẸP 1X21- 431 BT: 22.64 m (Tiêu hao: 23.52 m)
3	MDF-104_SH_-_6mm	2440.0 x 1220.0 x 6.0	2	6	
		Tổng	14	108	NẸP 1X21- 431 BT: 22.64 m (Tiêu hao: 23.52 m) N1X44-389: 10.21 m (Tiêu hao: 10.49 m) N1X21-389: 213.72 m (Tiêu hao: 229.52 m)

Thống kê gia công

#	Hạng mục	Số lượng
1	Độ dài dán cạnh cong	0.56 m
2	Số lượng bản lề	18
3	Số lượng ốc cam	82
4	Rãnh xoi 6x3	6.45 m
5	Rãnh xoi 7x7	5.00 m
6	Rãnh xoi 7x9	2.65 m

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